

Hantavirus Data – Descriptive Statistics Analysis

1 Introduction

This project applies descriptive statistics concepts learned from a textbook to a real hantavirus dataset from Kaggle, as a practice project.

Data source: <https://www.kaggle.com/datasets/zkskhurram/hantavirus-andes-virus-global-epidemiology>

Disclaimer: This is a historical Kaggle dataset on hantavirus epidemiology, not the 2025 outbreak data. This project focuses on learning descriptive statistics, not on current events.

2 Data Cleaning

No data errors were found. Clinical symptoms have structural missingness by syndrome. All outliers (Table ??) were genuine extreme values, not errors.

3 Univariate Distribution Analysis

3.1 Incubation Days

Right-skewed (skew=0.47), unimodal, thin tails. Mean=20.4d, Median=19d. Not normal. Incubation differs by syndrome: HPS (median=16d) vs HFRS (median=22d).

3.2 Hospital Days

Appears bimodal when aggregated – an artifact of mixing four severity groups with distinct non-overlapping centers (Mild=2d, Moderate=6d, Severe=13d, Critical=21d).

3.3 ICU Days

Spike-and-slab: 50% never admitted. Mild/Moderate never go to ICU; Severe/Critical always do. Among admitted: median=5d.

3.4 Environmental Variables

Temperature is approximately normal (skew=-0.15). Rainfall is right-skewed (skew=1.80), 10x variation by biome. Rodent abundance is symmetric and constant across biomes.

3.5 Confirmed Cases

Extreme right skew (skew=2.82). Mean=1793, Median=58. Tail driven by China HFRS.

3.6 Case Fatality Rate

Bimodal by syndrome. HPS (28.1%) is 20x more lethal than HFRS (1.4%).

4 Comparative Analysis

Incubation is similar across severities but differs by syndrome. Hospital/ICU days separate cleanly by severity. Cases and CFR differ dramatically by syndrome.

5 Key Takeaways

1. Check subgroups before describing a distribution
2. For skewed data: median + IQR, not mean + std
3. Spike-and-slab distributions need split analysis
4. Mixing groups creates misleading aggregates
5. Bar charts with error bars hide the truth

6 Tables

